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Microemulsion Surfactant Improved Performance and Cost of Single-Phase Low Viscosity Acid Systems

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Abstract

A significant portion of global oil and gas reserves are found in carbonate reservoirs, primarily composed of calcite and dolomite. To enhance production from these reservoirs, acid stimulation technologies are used to create conductive fractures or wormholes. Recently, we reported the use of single-phase low viscosity acid systems (LVAS), which is a mixture of concentrated organic and inorganic acids. However, the main challenge associated with LVAS is its higher cost when compared to emulsified acid.

A microemulsion surfactant, designed to improve flowback characteristics, achieved a desired retardation effect on acid and rock reaction rates. The properties, such as surface tension, stability on dilution and emulsion testing of the acidizing fluid containing acid retarder, were determined at a concentration of 2 gpt. To evaluate the performance of the acid, static dissolution testing was performed followed by corrosion evaluation at high temperature and finally characterization of acid propagation in core samples using a coreflood setup. Finally, CT scans of the acidized cores were undertaken to analyze wormhole propagation.

Static dissolution experiments were performed using calcite cores with acid formulations containing different ratios of organic acid to hydrochloric acid (HCl), both with and without the retarder. The corrosion rate was measured using T-95 metal coupons. Core flow tests were performed at 300°F, comparing the original and modified LVAS formulations to ascertain the volume of acid needed to achieve breakthrough. The acid retarder (FBE-R) was developed from a blend of ester or terpene solvents, alcohols, and surfactants. The modified formulation-maintained characteristics such as reaction rate with calcite, acid propagation and acceptable corrosion performance with tubing metallurgies. The addition of acid retarder (FBE-R) to the modified LVAS acid system (80/20 of 28 wt.% HCl / 70 wt.% MSA) enhanced its performance while reducing the volume fraction of the organic acid and also maintaining a corrosion rate below 0.05 lb./ft². Acid penetration rate into the core achieved similar volume of acid to breakthrough compared to the LVAS acid system. In addition, the reduction of 60 % (from 400 gpt to 160 gpt) in the required volumes of MSA would result in significant cost reduction in the acid stimulation cost.

The utilization of the new microemulsion surfactant, in conjunction with the LVAS acid system, has led to a significant reduction in the volume of MSA, decreasing the amount needed from 400 gpt to around 160 gpt. Such economic efficiency will support the wide implementation of the LVAS acid in the field to stimulate oil and gas wells drilled in carbonate reservoirs.

Introduction

The global energy demand continues to grow, driven by both renewable and non-renewable sources, and is expected to rise for decades. As reported by the United States Census Bureau, the world population reached 7.3 billion by mid-2015, marking an increase of one billion births in the previous twelve years (United Nations, Department of Economic and Social Affairs, Population Division, 2015). Projections suggest that by 2050, the population will expand to 9.7 billion, further intensifying the demand for energy. The three most populous countries—China, India, and the United States—are experiencing economic diversification and improving living standards, reinforcing the upward trend in energy consumption.

The oil and gas industry is well-positioned to tackle growing energy demands through continuous advancements in oilfield technologies. Innovations in smart chemicals and enhanced tools are enabling more efficient, cost-effective, and environmentally conscious production methods. According to [Schlumberger Market Analysis \(2007\)](#), carbonate reservoirs hold approximately 60% of global oil reserves and 40% of gas reserves. The mineral composition of these formations predominantly includes calcite and dolomite, often requiring acid stimulation for production enhancement ([Harris, 1961](#); [Abrams et al., 1983](#); [Fredd & Fogler, 1997](#); [Al-Anazi et al., 1998](#); [Mohamed et al., 1999](#); [Nasr El Din et al., 2000](#); [Frenier et al., 2001a, 2001b](#); [Kasza et al., 2006](#); [LePage et al., 2011](#); [Mahmoud et al., 2011a, 2011b](#)).

Acidizing techniques, such as matrix acidizing and acid fracturing, are widely applied based on specific reservoir characteristics. In acid fracturing, fluids are injected at high pressure to create etched fractures which sustain reservoir connectivity ([Al-Enezi et al., 2015](#)). Matrix acidizing, on the other hand, injects acid at lower pressures to form wormholes, which penetrate past the damaged zone, improving permeability without fracturing. These wormholes can extend from a few inches to several feet, significantly enhancing flow paths for hydrocarbon extraction.

Oil and gas operators utilize various acid platforms to stimulate carbonate formations, including strong mineral acids (e.g., hydrochloric acid), gelled and emulsified acids, and organic acids like formic and acetic acid ([Buijse et al., 2004](#); [Chang et al., 2008](#); [Gomaa & Nasr El Din, 2006 and 2008](#); [Navarrete et al., 1998a and b](#); [Rabie et al., 2012](#)). These methods enhance production but also present challenges affecting long-term viability.

To address these limitations, the industry seeks alternative acid systems that offer cost-effectiveness, stability, efficiency, biodegradability, low toxicity, and minimal formation damage. Additionally, controlled acid delivery is being explored to optimize fracture length and improve production rates.

Emulsified acids are widely used in carbonate stimulation due to their ability to slow reaction rates and promote efficient wormhole formation ([Buijse & van Domelen, 2000](#)). However, their high viscosity leads to significant friction loss, posing a challenge. Organic acids such as acetic and formic acid were introduced to mitigate metal corrosion in wellbores ([Van Domelen & Jennings, 1995](#)). Despite their benefits, these acids face drawbacks like high cost, weak dissolving power, and precipitation issues, which may cause formation damage ([Chatelain et al., 1976](#); [Chang et al., 2008](#)).

Recently, a low-viscosity retarded acid system (LVAS) was developed for carbonate stimulation, combining hydrochloric acid (HCl) for high dissolving power and strong organic acid (methane sulfonic acid, MSA) for slow reactivity, low corrosion rate, and improved handling. The 1:1 ratio of HCl and MSA in LVAS balances performance and formulation cost, as MSA is significantly more expensive compared to HCl. Compared to standard emulsified acid, LVAS is more expensive, even when factoring in that LVAS requires less volume to propagate a wormhole or create a fracture compared to emulsified acid system ([Sayed et al. 2018; 2020](#); [Majid et al. 2023](#)).

The application of water-based acidizing fluids in stimulating tight gas carbonate reservoirs often results in fluid retention challenges, which significantly influence the effectiveness of the pay zone. In acid fracturing operations, the byproducts of these reactions, commonly referred to as spent acid, tend to become trapped both within the fracture and in the surrounding rock ([Penny et al., 2005](#)). This issue is particularly

pronounced in low-permeability formations, where studies suggest that between 60% and 90% of the injected fluid may remain in the reservoir. Such retention can negatively impact relative permeability and fracture conductivity, reducing well productivity (Cuiec, 1989; Longeron et al., 1998).

This reduction in performance is attributed to increased water saturation near the wellbore, caused by the infiltration of aqueous fluids into gas-bearing zones. This process can lead to clay swelling, enhanced formation ductility, and a subsequent decline in rock permeability. When HCl at a concentration of 28 wt.% is used, it generates a high concentration, high density calcium chloride (CaCl_2) brine, which is particularly difficult to recover from tight carbonate formations. The challenge is exacerbated in low-pressure or depleted reservoirs, where fluid recovery becomes even more limited. During production, the movement of gas through these low-permeability zones may take a prolonged period to displace the retained fluids (Audibert-Hayet & Dalmazzone, 2006).

To displace imbibed fluids from porous media, the opposing force must exceed the capillary pressure. Higher capillary pressures make it more difficult for fluids to exit the pore spaces, leading to increased fluid retention within the formation. Capillary pressure can be mitigated by either lowering the surface tension (γ) or increasing the contact angle (θ), which involves altering the wettability of the rock. Consequently, various chemical additives have been explored to enhance the recovery of fracturing fluids (Kaufman et al., 2008; Cawiezel et al., 2010). Surface-active agents, including emulsion-based flowback enhancers (FBEs), are commonly employed in tight formations to facilitate the removal of retained fluids. These additives typically comprise blends or emulsions of surfactants and solvents designed to reduce interfacial and surface tension, thereby decreasing capillary pressure and minimizing water blockage.

The importance of such additives has grown with the rising focus on hydrocarbon extraction from low-permeability reservoirs (Mahmoudkhani et al., 2015; Kim et al., 2016). However, selecting an appropriate FBE for acidizing operations is complex due to the need for chemical stability and compatibility with 28 wt.% HCl and high-salinity brines under elevated temperatures. FBEs are not universally applicable and must be tailored to specific conditions, considering factors such as the composition of the treatment fluid, the mineralogy of the formation (e.g., sandstone, carbonate, or shale), and the type of reservoir fluid (oil or gas). Our development efforts have focused on micro/nano-emulsion-based FBEs, which have demonstrated superior performance compared to traditional surfactant-based micellar systems in enhancing flowback efficiency (Yue et al., 2016; Himes et al., 2017; Nelson et al., 2018; Saini et al., 2022).

As said, an in-house developed microemulsion surfactant that can work as a flowback enhancer and acid retarder (will be denoted as FBE-R) was developed and added to the LVAS formulation. The FBE-R was found effective at improving flowback properties of the spent acid as well as achieving retardation effects between acid and carbonates. The acid retarder/flowback enhancer additive was stable in 28 wt.% HCl at temperatures up to 325°F and was compatible with other acid additives. The rate of fluid flowback recovery (spent acid) and formation regained permeability both increased when the FBE-R was added.

The current work focuses on the efforts made to reduce the cost of the single-phase acid system. This was done by developing a new micro-emulsion and a double functioning acid retarder/flowback enhancer, to reduce the volume of the MSA and increase the volume of HCl while maintaining the same performance of LVAS acid systems. FBE-R is a micro-emulsion surfactant that was developed in-house and evaluated with acid systems and proved to be stable under reservoir conditions. The properties such as surface tension, stability on dilution and emulsion testing of the acidizing fluid containing acid retarder were all tested using a concentration of 2 gpt. To evaluate the performance of the acid, static dissolution testing was performed followed by corrosion evaluation at high temperature and finally coreflood testing to characterize acid propagation in core samples. Lastly, CT scans of the acidized cores were undertaken to analyze wormhole propagation.

Experimental Study

Materials

Unless otherwise noted, all starting materials were received from the vendor and used without further purification. A water purification system having a resistivity of 18.2 MΩ.cm was used to generate the de-ionized water (DI-H₂O) for all experimental procedures. A 37 wt.% HCl was purchased from local supplier and used to prepare the acid formulation. Methanesulfonic acid (70 vol%) initial concentration was purchased from local supplier, diluted and added to formulate the single-phase acid system (LVAS). In-house developed corrosion inhibitor (F-25) and an inhibitor intensifier, potassium iodide (KI), were added to the acid formulation to provide corrosion protection of metal tubing exposed to the single-phase acid (LVAS) fluid system. The iron control additives, i.e., L041 (tetrasodium ethylenediaminetetraacetate, Na₄EDTA) and L058 (sodium erythorbate) were received and added to the acid formulation.

A microemulsion surfactant (FBE-R) that can work as an acid retarder and flowback enhancer was developed in-house and evaluated with the new acid formulation. The FBE-R composition is shown below in Table 1. The sequence of chemical addition is in the order DI- H₂O, then alcohol, surfactant, and finally the solvent fluid. All chemicals were blended at high speed for a duration of 30 minutes until all components became miscible. The last component added was the D-limonene and blending continued an additional 10 minutes until a clear solution was obtained. D-limonene was used as an oil phase to form the oil-in-water microemulsions.

Table 1—Composition of the microemulsion acid retarder/flowback enhancer (FBE-R) used to retard acid reaction rate of the new acid system.

Formulation of Acid Retarder (FBE-R) Addition Sequence	Chemicals	Wt. %
1	De-ionized Water	45%
2	Propylene glycol	10%
3	Alkyl ether sulfate ammonium salt	30%
4	Secondary alcohol ethoxylate (EO)7	10%
5	D-Limonene	5%

Homogenous Indiana limestone core samples measuring 1.5" in diameter and 6" in length with permeability in the range of 1 to 5 mD were purchased from a local supplier in Texas USA. Cores were received and examined to determine mineral composition as well as characterizing the rock porosity and permeability of each core sample.

Methods

Preparation of The Acid Systems: In a typical experiment, the iron control additives and corrosion inhibitor intensifier were first dissolved in a beaker containing a pre-determined volume of DI- H₂O, i.e., the aqueous phase which will be later used to dilute the HCl to the desired concentration of 32 wt.%. Accordingly, the chelant (i.e., 0.6 wt.%; Na₄EDTA), iron stabilizer (i.e., 0.6 wt.%; sodium erythorbate), corrosion inhibitor intensifier (KI; 70 ppt) and the FBE-R (2 gpt) were added and fully dissolved in the DI- H₂O phase. A pre-determined volume of 37 wt.% HCl was then added to the DI- H₂O phase to yield a solution consisting of 32 wt.% HCl. Next, an equivalent volume of MSA (received as 70 vol%) was added to the beaker containing the 32 wt.% HCl solution (i.e., 1:1 ratio between MSA and 32 wt.% HCl). Lastly, the corrosion inhibitor (2 vol%; F-25) was added and the mixture permitted to stir until a homogeneous mixture was obtained.

Procedures for Core Preparation: The porosity for all Indiana limestone core samples used in this study was calculated using the saturation method. The as received 6" cores were first numbered sequentially and dried at 300°F (150°C) for at least 24h, cooled and weighed. A transparent protective coating (i.e., heat-shrink Teflon wrap) was then applied to each core. The final mass was recorded, and all core samples were subsequently saturated using DI- H₂O under vacuum for at least 24 hours. The saturated mass was recorded and used to determine the pore volume (cm³), which can be expressed as porosity (i.e., pore volume / bulk pore volume *100). Core samples having an average porosity of around 15.0 to 16.0% were selected for this study. The absolute permeability for each DI- H₂O saturated core sample was measured in a horizontal fashion using a high temperature, high pressure (HT/HP) coreflood apparatus. The permeability was calculated by flowing DI- H₂O through the core sample at various flow rates (e.g., ranging from 0.5 to 3 mL/min) until the flow stabilized. For each rate, the average differential pressure across the core (DP) was recorded and applied to Darcy's equation to determine the initial permeability. Those cores having permeability in the range of 1.0 to 5.0 md were selected for this study.

Characterization

Powder X-ray Diffraction: Powder X-ray Diffraction (PXRD) analysis verified the homogeneity of the Indiana Limestone core sample before acidizing, showing strong alignment between experimental and calculated diffraction patterns. The core's composition was calcite (99%), with minor amounts of quartz, dolomite and ankerite.

Corrosion Loss Experiments: A custom-designed autoclave (Paar reactor) was utilized to assess corrosion loss in the single-phase acid system (LVAS). The 400-mL reactor, rated for 400°C and 1900 psi, exposed carbon steel coupons to the acid system at 250°F and 275°F for a predetermined period of time. The acid system contains a corrosion inhibitor package matching what is utilized in the acidizing treatment jobs. Acceptable corrosion loss was set at <0.05 lb./ft². Before testing, coupons were preserved in mineral oil, cleaned with acetone, and carefully measured pre- and post-experiment.

Coreflood Experiment: For coreflood experiments, an HP/HT coreflood apparatus was utilized to simulate matrix acidizing conditions. The core flood setup is equipped with Hastelloy HC-276 wetted components which are acid resistant. The system operates under conditions up to 400°F and 10,000 psi. Back pressure, confining pressure, and pore pressure were maintained via Teledyne ISCO D100DX syringe pumps. Pressure drop across the core was tracked through Rawson pressure transducers and a Foxboro DP transmitter with a 5800 psi static pressure rating. Effluent samples were collected in 10 cc increments using a Teledyne Retrieve 500 Auto-sampler. Fig. 1 shows a schematic for the core flood setup used in the current study.

Inductively Coupled Plasma (ICP) Measurements: Calcium concentration in effluent samples from coreflood acidizing experiments was quantified in mg/L (ppm) using Inductively Coupled Plasma Optical Emission Spectrometry (ICP-OES). Samples were serially diluted (100 times) and assessed against a linear calibration curve ranging from 0.00 to 1,000 mg/L. Calibration verification standards were measured every 12 samples to ensure accuracy. Each reported value represents three replicates, measured at an injection rate of 2 mL/min for 6 minutes.

X-ray Computed Tomography (CT) Imaging & Analysis: To assess acid propagation behavior, core samples were imaged using an industrial X-ray tomography CT scanner. It features a maximum resolution of approximately 500 nanometers, enabled by geometric magnification greater than 3000x and X-ray energies ranging from 10 kV to 450 kV. The system supports both 2D digital radiography and 3D computed tomography, making it ideal for detailed internal inspections across a wide range of industries.

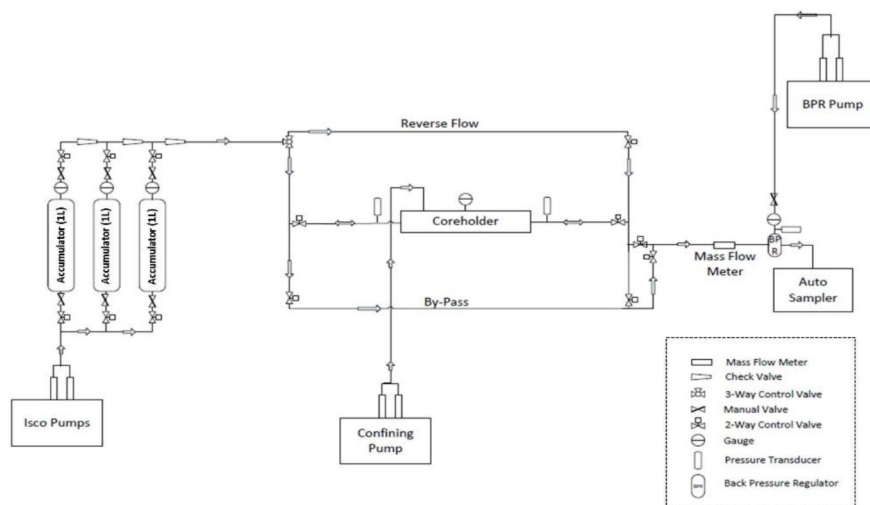


Figure 1—Schematic of the custom-designed coreflood set-up used to perform the experiments outlined in this section.

Results and Discussions

Development of the Acid Retarder/Flowback Enhancer (FBE-R)

Several flowback enhancer formulations were developed and evaluated for their potential use in acid stimulation treatments aimed at improving the efficiency of carbonate gas reservoir stimulation. Key performance criteria for these additives included their thermal stability in acidic environments at temperatures reaching up to 300°F, as well as their efficiency in both live and spent acid as well as their impact on the reaction rate of calcite rocks and acid systems.

Preparation of the Acid Retarder Flowback Enhancer (FBE-R): The newly developed formulations were created by combining different surfactants, alcohols, and solvents into aqueous mixtures, resulting in oil-in-water microemulsions. The FBE-R was engineered to exhibit zwitterionic properties. Zwitterionic surfactants were specifically selected due to their strong resistance to high-salinity environments, particularly in the presence of multivalent ions. These surfactants also offer excellent thermal stability and are less likely to adhere permanently to the positively charged surfaces of carbonate formations. Zwitterionic surfactants behave like cationic agents in acidic conditions but include a countercharge in their molecular structure, enhancing their solubility in aqueous environments. Additionally, these surfactants can generate foam in high-density brines, which is beneficial for unloading fluids in low-pressure gas reservoirs. The inclusion of thermally stable nonionic ethoxylates in the formulations helps prevent emulsion formation with hydrocarbon condensates and contributes to lowering surface tension.

Droplet Size Characterization: The FBE-R developed in this study appeared as clear liquids, which is indicative of successful formation of microemulsions. To verify the droplet size distribution, measurements were conducted using a Malvern Zetasizer in both 2 wt.% KCl and 28 wt.% HCl solutions. The results showed an average droplet sizes of 507 nm and 172 nm in 2 wt.% KCl and 28 wt.% HCl, respectively, ranging from 11 to 802 nanometers, confirming the presence of oil-in-water nano- and microemulsion structures.

Stability in Acidizing Fluids: The stability of FBE-R was assessed in 28 wt.% HCl at concentrations of 2 and 5 gpt. Surface tension measurements were conducted using a Ramé-Hart goniometer tensiometer via the sessile drop method. These measurements were taken both immediately after the FBE-R was mixed with the acid and again after the solutions were aged for 24 hours at ambient temperature and pressure. As summarized in Table 2, only minor variations in surface tension were observed across the formulations, suggesting that the surfactants maintained stability under acidic conditions at room temperature.

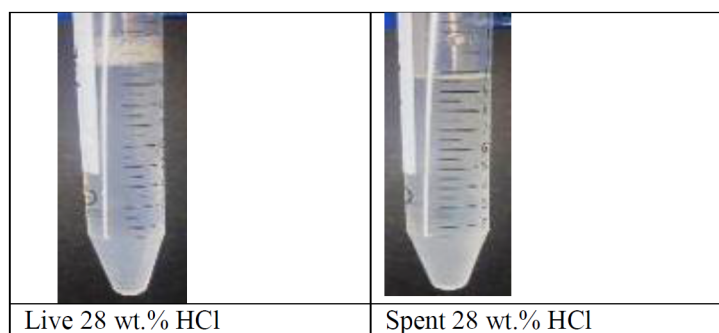
Table 2—Surface tension (mN/m) of FBE-R in 28 wt.% HCl (fresh and spent) measured at room temperature and ambient pressure.

Temperature and Test Duration	28 wt.% HCl/2 gpt FBE-R	Spent 28 wt.% HCl/2 gpt FBE-R	28 wt.% HCl/5 gpt FBE-R	Spent 28 wt.% HCl/5 gpt FBE-R
Room Temperature	23.7	27.8	21.8	25.9
250°F (24 h)		28.0		
325°F (24 h)		29.8		

Further testing was carried out under simulated spent acid conditions to replicate the high-density brine environment typically encountered during production (Table 2). Under these conditions, there was no significant change in the surface tension measurement indicating that the FBE-R was stable in spent acid formulation with high concentration of calcium ions.

To assess thermal stability under reservoir conditions, surface tension measurements were conducted at temperatures of 250°F and 325°F. This involved heating a mixture of spent 28 wt.% HCl with a 2 gpt of FBE-R in a pressurized cell for 24 hours. After the heating period, the fluids were cooled, and their surface tension values were recorded (Table 2) and compared to the pre-heating values. The results showed minimal changes in surface tension indicating that these formulations remained stable and compatible under high-temperature conditions.

The next step was to visually inspect the solution of acids up on the addition of the FBE-R. Additionally, any solid formation within the acidizing fluid or brine (spent acid) could cause plugging to the pore spaces and reduce the permeability of the reservoir rock. To evaluate stability of the FBE-R, a concentration of 2 gpt was added to both live and spent 28 wt.% HCl. Each prepared solution was stored in a vial at ambient conditions for 24 hours to monitor for signs of phase separation or precipitation. As shown in Fig. 2, no visible separation or solid formation was observed indicating the stability of the developed FBE-R in both live and spent acids.

**Figure 2—Stability of FBE-R in both live and spent 28 wt.% HCl (left).**

Spent Acid Fluid Displacement/Recovery Column Test with Gas: To replicate fluid recovery from carbonate reservoirs using gas flow, columns filled with carbonate particles (400 μm) were used to conduct fluid displacement testing using nitrogen. A 28 wt.% HCl spent acid solution without FBE-R additive was tested to establish a baseline for fluid recovery. The results from this test were then used to assess the impact of incorporating FBE-R into the fluid. The baseline test, using only the spent acid without FBE-R, achieved a fluid recovery of 16% in a time period of 60 minutes. However, the inclusion of 2 gpt of FBE-R significantly enhanced fluid recovery (71%) (Fig. 3). When the concentration of the FBE-R increased to

5 gpt, the amount of spent acid recovery increased to 81%. Based on these findings, FBE-R was selected as a promising candidates for use in acidizing treatments.

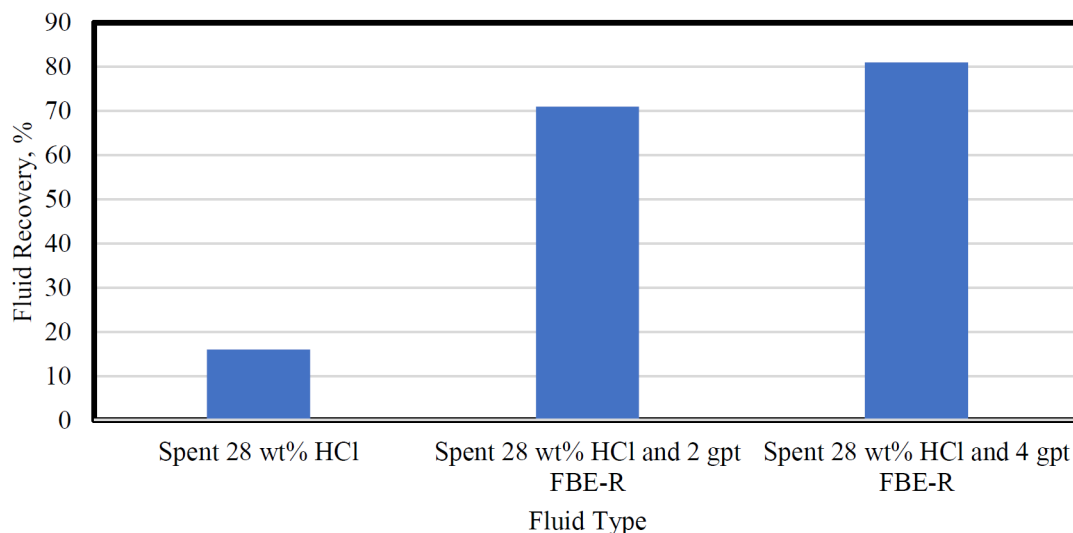


Figure 3—Fluid recovery of spent 28 wt.% HCl fluid with and without FBE-R.

The Impact of Adding the FBE-R on Acid Dissolving Power

A series of benchtop dissolution tests were initially conducted under static and room temperature conditions to identify the impact of adding the FBE-R to the acid fluid dissolution of calcite rock in acid and hence the retardation effect. We started with testing the FBE-R with HCl fluids. After the determination of the ability of the FBE-R/HCl-based fluids, the FBE-R was added to the single-phase acid system (LVAS) to investigate any incompatibility issues and to investigate the dissolution of the calcite rock in the acid formulation. In each independent experiment, the as-prepared Indiana limestone core sample was exposed to the fluid for five minutes. The core sample was subsequently removed from the beaker containing live acid and immediately washed with DI-H₂O to remove any residual acid. The acidized core sample was then dried, and the weight recorded to determine the percentage of calcite dissolved by mass.

The addition of FBE-R to the acid formula increased its retardation as indicated by the reduction in the amount of rock dissolved in the acid. The first screening tests were performed by adding the FBE-R to a straight HCl acid system (28 wt.%) just to study the compatibility as well as the effect on the dissolution rate of calcite in acid. A summary of the dissolution data is summarized and shown in Table 3. As shown in Table 3, the amount of limestone dissolved decreased from 21.1% to 16.8% as a result of the addition of FBE-R at a small concentration of 2 gpt to the 28 wt.% HCl. As a result, the utilized micro-emulsion FBE-R has the potential to enhance the performance of the single-phase acid system (LVAS) if added to the formula.

Table 3—Effect of FBE-R in retarding HCl reactivity in limestone.

Chemical / Description	Test #1		Test #2	
	gpt	mL	gpt	mL
36 wt.% HCl	752.4	188.1	752.4	188.1
DI- H ₂ O	165.6	41.4	163.6	40.9
F25, in-house CI	20	5	20	5
CI Intensifier, A201	60	15	60	15
L041, chelating agent	50 pptg**	1.5 g	50 pptg**	1.5 g
L058, reducing agent	4 pptg**	0.12 g	4 pptg**	0.12 g
M295, H ₂ S scavenger	2	0.5	2	0.5
FBE-R, acid Retarder	0	0	2	0.5
TOTAL	1000	250	1000	250
% Dissolved (by Mass)	21.1		16.8	

**pptg Pound Per Thousand Gallons of Formulation

Next, the impact of the addition of FBE-R to the LVAS acid system was evaluated at room temperature and under static conditions. The data are summarized and shown in [Table 4](#). The addition of FBE-R at a concentration of 2 gpt resulted in the reduction of mass of rock dissolved from 15 to 11%.

Table 4—Static dissolution testing performed under ambient conditions to evaluate the effect of FBE additive on rock dissolution for LVAS system.

Chemical / Description	Test #3		Test #4	
	gpt	mL	gpt	mL
31 wt.% HCl	400	100	400	100
DI-H ₂ O	178	44.5	176	44.0
methanesulfonic acid (MSA)	400	100	400	100
F25, in-house CI	20	5	20	5
CI Intensifier, KI	70 pptg	2.1 g	70 pptg	2.1 g
L041, chelating agent	50 pptg	1.5 g	50 pptg	1.5 g
L058, reducing agent	4 pptg	0.12 g	4 pptg	0.12 g
M295, H ₂ S scavenger	2	0.5	2	0.5
FBE-R, acid Retarder	0	0	2	0.5
TOTAL	1000	250	1000	250
% Dissolved (by Mass)	15.1		11.7	

Formulation Optimization: Selection of Suitable Acid Ratios

The next step was to investigate the addition of FBE-R to LVAS acid systems while changing the volume ratio of both organic and mineral acids. The increase in inorganic/organic acid ratio (28 wt.% HCl to 70 wt.% MSA) from 60/40 to 80/20 yielded a subtle effect on calcite dissolution. It was observed that the increase in the ratio of 28 wt.% HCl to the MSA resulted in an increase in the dissolution of calcite in the acid system.

However, the acid composition will affect the economics of the treatment. Therefore, retarding the dissolution of calcite in acid specifically for these formulations by the addition of FBE-R was attempted as shown in [Table 5](#). The percentage of dissolved calcium carbonate dropped from 15.1 for the original LVAS acid system to 10.35% when adding FBE-R at a concentration of 4 gpt to an acid system that has 80/20 of 28 wt.% HCl to MSA (more HCl and less MSA formulation). The addition of FBE-R at a concentration of 2 gpt to the modified LVAS formulation that has an 80/20 of 28wt% HCl to MSA achieved the same dissolution as the original LVAS formulation.

Table 5—Static dissolution testing performed under ambient conditions to evaluate the effect of FBE-R additive on rock dissolution for LVAS-1 system by reducing the concentration of MSA in the formulation.

Chemical / Description	Test#5 (Original LVAS)		Test #6 (80/20 HCl/MSA + 2 gpt FBE-R)		Test #7 + 4 gpt FBE-R	
	gpt	mL	gpt	mL	gpt	mL
31 wt.% HCl	400	100	640	160	640	160
DI-H ₂ O	178	44.5	176	44.0		43.5
methanesulfonic acid (MSA)	400	100	160	40	160	40
F25, in-house CI	20	5	20	5	20	5
CI Intensifier, KI	70 pptg	2.1 g	70 pptg	2.1 g	70 pptg	2.1 g
L041, chelating agent	50 pptg	1.5 g	50 pptg	1.5 g	50 pptg	1.5 g
L058, reducing agent	4 pptg	0.12 g	4 pptg	0.12 g	4 pptg	0.12 g
M295, H ₂ S scavenger	2	0.5	2	0.5	2	0.5
FBE-R, w	0	0	2	0.5	4	1
TOTAL	1000	250	1000	250	1000	250
% Dissolved (by Mass)	15.1		15.0		10.35	

Evaluation of Fluid Performance: Coreflow Analysis & CT Imaging

To evaluate the performance of this acid platform and gain a deeper understanding of the dissolution behavior, a series of systematic coreflood experiments were conducted in parallel with ICP analysis of the effluent samples where the calcium concentration was detected in order to calculate the amount calcite dissolved. To characterize the wormhole propagation behavior, the acidized cores were subsequently imaged using an industrial X-Ray tomography.

The coreflood experiment was performed at the following conditions:

- Flow rate: 1 mL/min
- Temperature: 300°F
- Back Pressure 3000 psi
- Confining Pressure of 4000 psi

LVAS Acid System - No FBE-R

LVAS acid system was prepared including all the formulation additives as advised by industry practices. The experiment started by injecting DI-H₂O to establish the initial pressure drop across the core. Then the acid was injected at the same injection rate of 1 mL/min. When acid was injected into the core, the measured differential pressure across the core started to increase as a result of the higher viscosity of the LVAS fluid. With more acid injected into the core, the differential pressure across the core started to decrease as the injected acid dissolved part of the rock creating wormholes as the acid penetrates deeper into the rock matrix. Eventually, acid breakthrough occurs, as indicated by the sudden drop in the differential pressure across the core (Fig. 4). Reverse flow with DI-H₂O from the opposite direction was established to recover the dissolved calcium carbonate for ICP analysis.

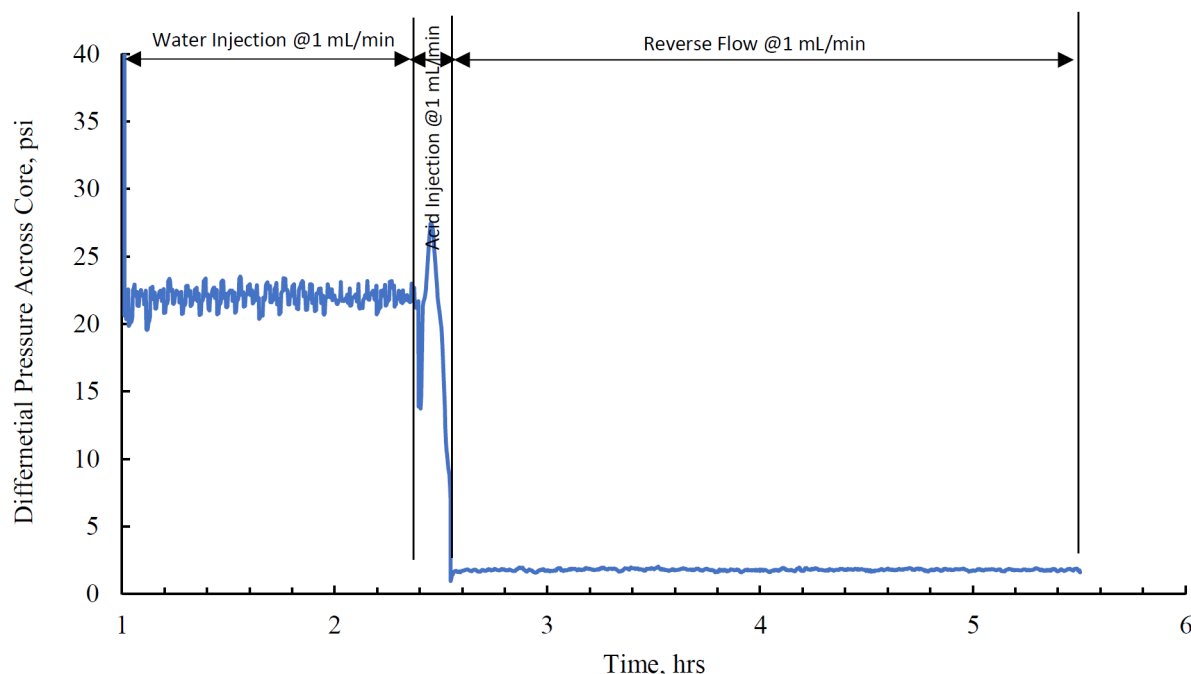


Figure 4—Pressure drop profile during injection of LVAS-1 at a rate of 1.0 mL/min.

A volume of acid equivalent to 0.34 pore volume was needed to achieve breakthrough or create a wormhole that extends the full length of the core sample. Acid volume to breakthrough (PVB_T) is the volume of acid needed to create a wormhole that extends the full length of the core samples expressed as pore volume (normalized).

Original LVAS Acid system Plus FBE-R (2 gpt)

The same procedure followed in the previous experiment was followed in the 2nd experiment. The fluid system used here was the same LVAS fluid with 2 gpt of FBE-R added. The differential pressure across the core was recorded and plotted as a function of time and shown as in Fig. 5. Acid injection continued until acid breakthrough was observed and then water was injected in the reverse direction to stop acid invasion and to simulate acid flowback. A volume of acid equivalent to 0.19 pore volume was needed to achieve breakthrough across the 6-inch core sample. It is obvious that the addition of the FBE-R to the LVAS acid system resulted in more retardation as reflected by the reduction in the acid volume needed to achieve breakthrough as it decreased from 0.34 to 0.19 PV.

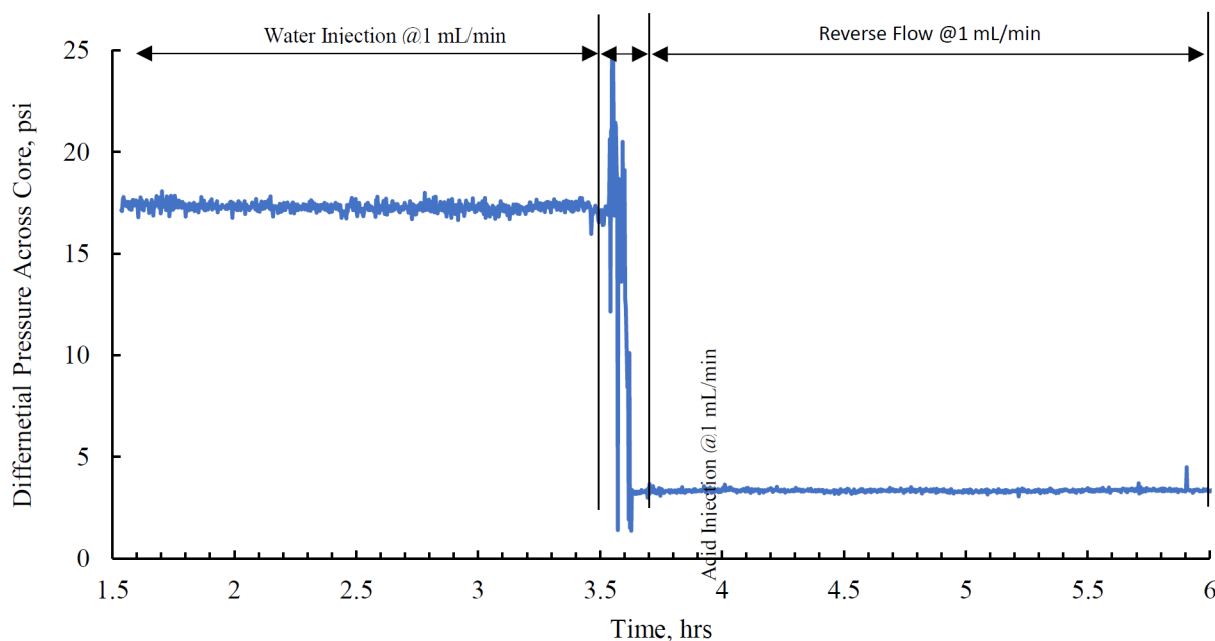


Figure 5—Pressure drop profile during injection of LVAS-1 + FBE-R (2 gpt) at a rate of 1.0 mL/min.

Modified LVAS-1 Acid system with and without the FBE-R (2 and 4 gpt)

A modified acid formulation was used and tested in this set of experiments (Run 3 to Run 5). The composition of the acid system is shown in Table 6. The acid formulation used here uses a ratio of 80/20 of 28 wt.% HCl and 70 wt.% MSA. The pore volume of acid needed to achieve breakthrough is shown in Table 6.

Table 6—Composition of the modified LVAS-1 acid systems used in the coreflood experiments.

Chemical / Description	Test #3		Test #4		Test #5	
	gpt	mL	gpt	mL	gpt	mL
31 wt.% HCl	640	160	640	160	640	160
DI-H ₂ O	176	44.0	176	44.0		43.5
methanesulfonic acid (MSA)	160	40	160	40	160	40
F25, in-house CI	20	5	20	5	20	5
CI Intensifier, KI	70 pptg	2.1 g	70 pptg	2.1 g	70 pptg	2.1 g
L041, chelating agent	50 pptg	1.5 g	50 pptg	1.5 g	50 pptg	1.5 g
L058, reducing agent	4 pptg	0.12 g	4 pptg	0.12 g	4 pptg	0.12 g
M295, H ₂ S scavenger	2	0.5	2	0.5	2	0.5
FBE-R, flow back enhancer	0	0	2	0.5	4	1
Acid Volume to Breakthrough, PV		0.373		0.23		0.226

The addition of FBE-R to the modified LVAS acid system (80/20 of 28wt% HCl / 70 wt.% MSA) enhanced its performance by retarding the acid reaction and enhancing acid penetration rate into the core sample. This was confirmed by the reduction of the acid volume to achieve breakthrough from 0.373 (for

the modified LVAS fluid system without FBE-R) to 0.23 (modified LVAS acid fluid plus FBE-R). This presents a 39% reduction in the acid volume needed to achieve breakthrough just obtained by adding FBE-R. Also, this indicates that the performance of the new formulation is similar to the performance of the originally used LVAS formulation as the volume of the acid to achieve breakthrough was reported to be 0.20 PV. Adding more FBE-R (4gpt) as shown in Run 5 did not show more enhancement compared to Run 4 (2 gpt FBE-R) as shown in Figure 6.

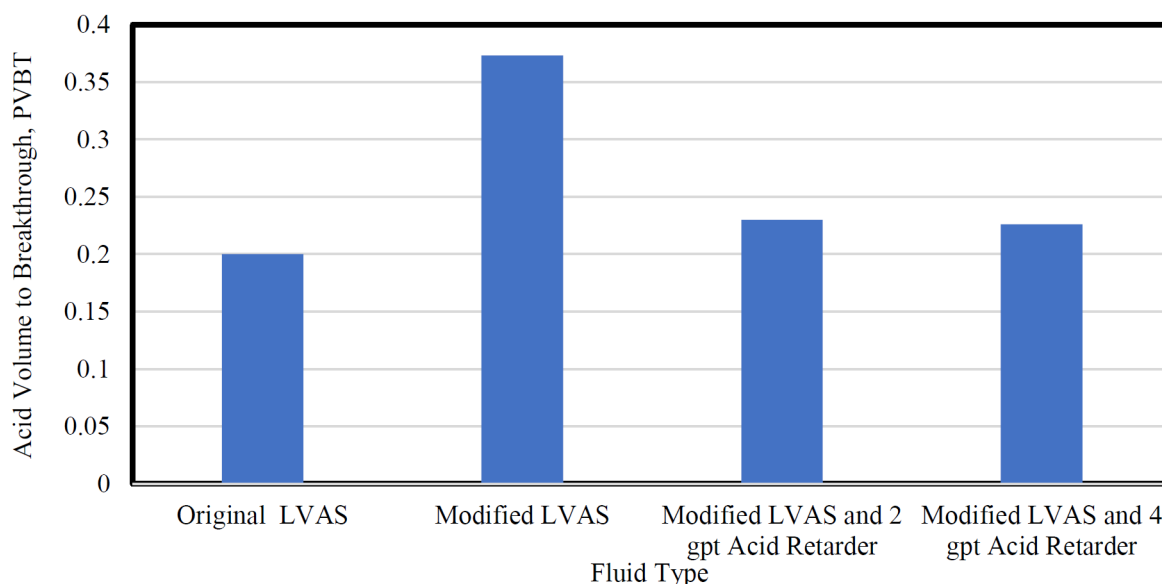


Figure 6—Acid volume to Breakthrough for modified LVAS-1 with and without FBE-R

Corrosion Loss Evaluation

The corrosion properties of the original single-phase LVAS acid system and the modified acid formulation (80/20 HCl/MSA) were measured using a high pressure, high temperature autoclave. The device was equipped with a magnetic drive, where the metal rod was used to secure the custom-designed Teflon coupon holder that could hold two metal coupons per experiment. In a typical experiment, the reservoir was filled with 250-mL of the acid fluid system. The Autoclave head, containing the metal coupons, was then secured. The temperature controller was programmed to increase the temperature at a steady rate until it reaches the set temperature. The duration of the test was varied depending on the temperature of the experiment. The test duration was around 3 and 6 hours for 275 and 250°F, respectively. The following procedure was followed in all experiments to have a solid base to compare results:

- Prepare 250 mL of the acid solution according to the specified formulation.
- Prepare fluids systems with and without the addition of FBE-R.
- Clean the metal coupons by rinsing them in acetone and gently brushing the surface with a brush to eliminate any residual oils. Ensure the coupons are thoroughly dried.
- Use an analytical balance to weigh each coupon to four decimal places.
- Measure the dimensions of the coupons to determine their surface area.
- Capture digital images of both sides of each coupon prior to testing.
- Attach the coupons securely to the designated holder.
- Transfer the acid solution into the reactor vessel and seal it with the cap.

- Pressurize the vessel to 1,000 psi using nitrogen gas and begin heating. The test is conducted under static conditions.
- The timer is started when the coupons enter the test fluid so that the test time will be equivalent to total contact time. About 30 minutes before the end of the test, turn off the heater to allow the vessel to cool to a safe temperature for coupon removal.
- After the vessel has cooled to room temperature, remove the coupons, rinse them with water followed by acetone, and gently scrub off any remaining residue. Let the coupons dry in air for approximately 2 minutes.
- Record the final weight of each coupon and take post-test photographs.
- Calculate the corrosion rate and assess visual pitting.

After calculating the total surface area, the corrosion rate (lbs./ft² @ time) is calculated using:

$$\text{Corrosion Rate (lbs./ft}^2\text{)} = \text{Average coupon loss (g)} \times \text{strip factor (lb./g}\cdot\text{ft}^2\text{)}$$

The corrosion loss data calculated for both acid systems as a function of varying temperature and exposure time is presented in Fig. 7. Table 7 shows pictures of the metal coupons before and after exposure to acid. The corrosion inhibitor package assessed in the current set of experiments was found to be effective to inhibit corrosion when combined with the new acid system (80/20 HCl/MSA + FBE-R). The corrosion loss measurement was less than 0.035 lb./ft² after soaking for 6h at 250°F.

Table 7—Coupons photos and Corrosion Rate for the two acid systems assessed in the study.

Fluid System	Temperature, °F	Test Time (h)	Coupon	Corrosion Rate
Original LVAS	250	6		0.0238
Modified LVAS + 2 gpt FBE-R	250	6		0.0356
Original LVAS	275	3		0.0298
Modified LVAS + 2 gpt FBE-R	275	3		0.0412

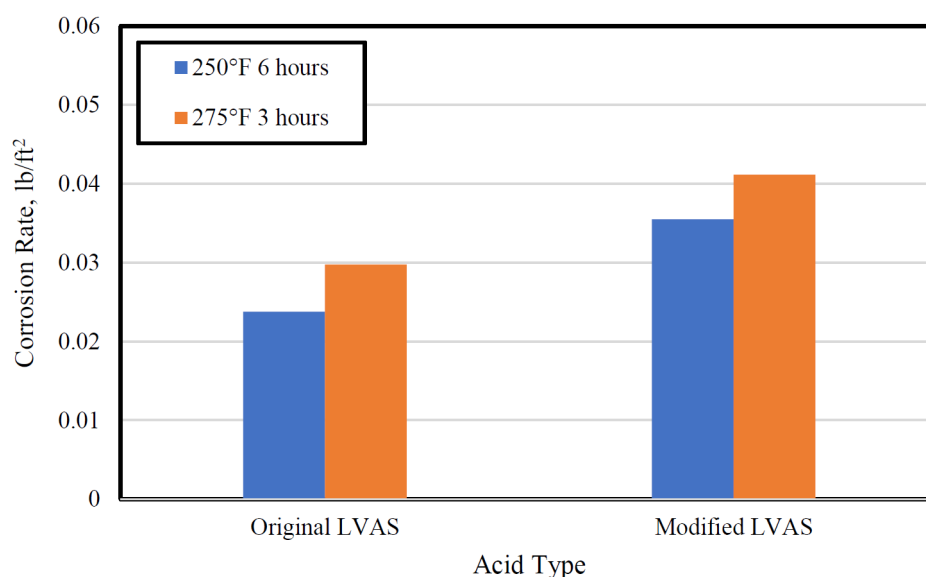


Figure 7—Summary of corrosion loss experiments for the two acid systems performed at temperatures of 250°F (6 hours) and 275°F (3 hours). Note that a corrosion loss less than 0.05 lb./ft² is considered acceptable by industry standard.

At a temperature of 275°F and after metal exposure to the acid fluid system for 3 hours, both acid systems achieved corrosion loss of less than 0.05 lb./ft² indicating that both acid systems met the criteria for corrosion loss mitigation. The addition of FBE-R was proven not to impact the performance of the corrosion inhibition package and corrosion loss targets were met in both cases.

Conclusions

1. A microemulsion surfactant was developed and evaluated as double acting acid retarder and flowback enhancer. The acid retarder/flowback enhancer was denoted FBE-R. The surface tension of the fluid system with the FBE-R additive remained stable after the spent acidizing fluid containing these additives was subjected to thermal aging at 250°F and 325°F for 24 hours.
2. In fluid displacement column tests in a carbonate-packed column using spent acid fluid with and without FBE-R, the fluid recovery increased from 16% (no FBE-R) to 71% and 81% for 2 and 5 gpt of FBE-R, respectively.
3. The dissolution experiments demonstrated that incorporating FBE-R into the acidizing fluid further slowed the reaction rate between the acid and the carbonate rock, even when all additives were present.
4. A new low-viscosity acid platform having the desired retardation properties and a repertoire of additional benefits was successfully developed, as exemplified by the design and characterization of new acid systems. The new acid system is a modification of the successfully deployed single phase acid (LVAS).
5. The modified formulation used less volume of methane sulfonic acid (MSA) and utilized the use of micro-emulsion surfactant (FBE-R) in a dosage of 2 gallons per thousand gallon of the treatment (2 gpt) to preserve the desired acid properties of reaction rate with calcite rich reservoirs, acid propagation rate and corrosion performance with tubing metallurgies.
6. The addition of FBE-R to the modified LVAS acid system (80/20 of 28wt% HCl / 70 wt.% MSA) enhanced its performance by retarding the acid reaction and enhancing acid penetration rate into the core sample as evidenced through the outcome of the coreflood experiments and how this system achieved similar volume of acid to breakthrough as compared to the original LVAS acid system.

7. The addition of the FBE-R did not impact the corrosion performance of the acid system and did not compromise the performance of the corrosion inhibition package where the corrosion rate measured was less than 0.05 lb./ft².
8. Finally, the volume of MSA was reduced from 400 gpt to around 160 gpt. This resulted in a substantial saving in the cost of the final acid formulation where an estimated cost saving of 40% can be achieved.

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